Applied Stats Paper

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The NBA is a professional basketball league that contain 32 teams that compete in an 82 game regular season. While the leage gives out several prestigious awards like Rookie of the Year, Clutch Player of the Year, Defensive Player of the Year, etc,; one the leagues greatest honors is the award for Most Valuable Player (MVP), whihc is given to the player who was deemed to have the greatest over impact based on performance, team success and media voting.

The main statistics used in this analysis are as follows:

- Points per game: the average number of points a player scores per game
- Assists: the average number of passes that lead to a teammate's basket per game
- Total Rebounds: the average number of rebounds per game, including both offensive and defensive
- True Shooting Percentage: a measure of scoring efficiency that accounts for field goals, three points shots and free throws
- Value Over Replacement Player (VORP): an advanced statistic used to estimate a player's overall contribution and on-court value
- Blocks: the average number of opponenet shot attemps a player blocks per game
- Steals: the average number of time a player manages to take the ball away from an opponent per game

The NBA MVP award is widely accepted as one of the highest individual honors in professional basketball, but it's important to acknowlege that its selection process is inherently subjective. Unlike the Golden Boot in professional soccer or the Brownlow Medal in the Australian Football League, which are awards based purely on statisitcs, the MVP award is determined by around 100 various sportswriters and broadcasters, meaning that their votes are influenced not only by measureable performance but an overall narratives.

A commonly mentioned phenomenon among NBA fans and even analysts is voter fatigue, which is used to describe the tendency for vtoesr to avoid awarding the same player consequtive MVPs. While voter fatigue has no official role in the voting criteria, it is widely belived to influence outcomes with many fans and analsts arguing that all-time great players, such as Michael Jordan, have lost out on MVP awards despite providing the best statisical performance.

In addition to voter fatigue, most MVP winners are sleected from teams with strong regular season records. This suggests that team success plays a signifcant role in the voters perception of the indiviidual palyer. This creates a bias that players on top performing teams may be favored over more productive players on weaker teams.

Finally, media narrative, such as a players storyline or their team leadership may influence the voting decisions. These factors highlights a disconnect between the MVP outcomes and the objective statistical performance. This paper aims to answer can a constructed statistical metric accurately identify NBA MVP-level players, and how well does it match actual MVP voting or are voters being significantly influenced by subjective considerations?

All data was collected from Basketball Reference, a widley used source for NBA statistics. This dataset consists of player level season averages where each row represents a sinlge player's performance during a single NBA regualr season. This analaysis looks at the past 11 seasons for a suficient sample size. It is important to note that the MVP voting occurs before the postseason begins, becuse of this, all post season performance was excluded from this analysis. This is further supported by Kirschner, a former MVP voter, at the The New York Times who states, "what happens in the playoffs is totally meaningless and irrelevant" in the MVP voting process.

For each player in each season, both tradisitional and advanced stastisics were colleced. The traditional statistics include points per game(PTS), assists(AST), total rebounds(TRB), steals(STL) and blocks(BLK). Additionally, advanced statistics such as True Shooting Percentage (TS%) and Value Over Replacement Player (VORP) were included. Combined, theses variables represent the ket aspects of offensive production, playmkaing, and defensive skill.

The dataset was created by downloading separate per-game and advanced stastiscs tables for each season and merging them by player name and season. In cases where players were traded mid-seasons, the "total" (TOT) row was used to represent thier full season performance.

Clean and organize data

```python
# Read in and Merge Data

import pandas as pd

seasons = [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025]

all_seasons = []
```

```
for season in seasons:
  #get file name for each season
  per_game_file = f"data/per_game_{season}.csv"
  advanced_file = f"data/advanced_{season}.csv"

  #load in files
  per_game = pd.read_csv(per_game_file)
  advanced = pd.read_csv(advanced_file)

  #clean header rows
  per_game = per_game[per_game["Rk"] != "Rk"].copy()
  advanced = advanced[advanced["Rk"] != "Rk"].copy()

  #clean players who were traded
  per_game = per_game[(per_game["Team"] =="TOT") | (~per_game["Player"].duplicated(keep=Fals
  advanced = advanced[(advanced["Team"] =="TOT") | (~advanced["Player"].duplicated(keep=Fals

  #create season column
  per_game["Season"] = season
  advanced["Season"] = season

  #columns to keep, keeping extra columns to allow flexibility while creating formula
  advanced_keep = advanced[["Player", "Season", "TS%", "PER", "BPM", "VORP", "WS"]].copy()

  #merge data
  merged_nba = pd.merge(
    per_game,
    advanced_keep,
    on = ["Player", "Season"],
    how = "inner"
  )

  all_seasons.append(merged_nba)

#combine seasons in dataset
combined_nba = pd.concat(all_seasons, ignore_index = True)

final_nba = combined_nba[["Player", "Season", "Age", "Team", "G", "PTS", "AST", "TRB", "STL"
```

In any NBA analysis, there are limitations in the data. The first being that certain aspects of a players performance such as leadership, team context, and off-ball defesne are not captured by box score statistics; so while they are potentially relevent, the NBA currently has not way of quanitiatively measuring them. Secondly, advanced statsitical metrics, such as VORP, are model based estimates meaning that they may not perfectly reflect a player's true value.

To evaluate palyer performance, a MVP metric was created using a weighted combination of several statistics. This approach was used to allow for the integration of scoring, palymaking, defensive contribtions and efficiency into a single quantitative measure. The variables used include points per game, assists, total reobunds, true shooting percentage, VORP, and a defensive metric that combined steals and blocks. Because these statistics are all measured on different scales each variable was statndardized within each season using z-scores. Z-scores are important for explaining how far above or below the average a player is. To calculate a z-score, we use the following formula:

$$Z = \frac{X - \mu}{\sigma}$$

X = player's stat for the respective season

μ = league average of that stat for the respective season

σ = standard deviation of that stat for the respective season

This works because $X - \mu$ tells how far from the average a player is and dividing by the standard deviation, or σ allows for a proper scale.

The MVP Score was calculated as a weighted sum of the standardized statsitics, with a greater weight placed on scoring and overall impact.

$$MVPScore = 0.30 * PTS + 0.20 * AST + 0.10 * TRB + 0.15 * TS + 0.15 * VORP + 0.10 * DEF$$

The weights associated with each statistic were chosen to reflect the different aspects of player performance while maintianing a balance between offensive and defensive production, efficicency and overall impact. However, it is important to note that the weights themselces are somewhat subjective and are aimed at reflecting the commonly accepted priorites when evaluating player performance. Scoring was given the highest weight (30%) as points per game is the most direct measure or offensive production and is consisitenyl a key factor in MVP voting. Assists were also weight heavily (20%) to caputre a player's ability to generate offense for their teammates. Rebounding was given a smaller weight as getting or maintaining possesion is importnat, it is usually looked at less in MVP evaluation. True shooting percentage (15%) was added to account for socring efficiency, this checks that players are not being rewarded solely for high volumen shooting. Value over replacement palyer (15%) was included to capture their overlal player value that is not typically included in traditionaly box score statistics. Finally, a defensive component (10%) was added that combined steals and blocks to account for defnsive impact, while still undetsainding that there are limitations to measuring defesnce.

The weighted format was designed to emphasize offensive production while still incorporting their efficiency and defense.

Create MVP Score Metric

```
#make defense metric
final_nba["DEF"] = final_nba["STL"] + final_nba["BLK"]

#scale data - used nba_clustering activity as source
columns = ["PTS", "AST", "TRB", "TS%", "VORP", "DEF"]

for column in columns:
  final_nba[f"scaled_{column}"] = final_nba.groupby("Season")[column].transform(
    lambda x: (x - x.mean()) / x.std()
  )

final_nba["MVP_Score"] = (
  0.30 * final_nba["scaled_PTS"] +
  0.15 * final_nba["scaled_AST"] +
  0.10 * final_nba["scaled_TRB"] +
  0.15 * final_nba["scaled_TS%"] +
  0.20 * final_nba["scaled_VORP"] +
  0.10 * final_nba["scaled_DEF"]
)
```

Find actual MVP

```
final_nba["Rank"] = final_nba.groupby("Season")["MVP_Score"].rank(ascending = False)

final_nba[final_nba["Rank"] <= 5 ].sort_values(["Season", "Rank"])

# create indidivual data frames for each season for better output
for season in final_nba["Season"].unique():
  print(f"\nSeason {season}")
  display(
    final_nba[
      (final_nba["Season"] == season) &
      (final_nba["Rank"] <= 5)
```

```
    ][["Rank", "Player", "MVP_Score"]].sort_values("Rank")
  )
```

```
Season 2015
```

	Rank	Player	MVP_Score
0	1.0	Stephen Curry	3.304772
2	2.0	Kevin Durant	2.850037
1	3.0	James Harden	2.814200
8	4.0	Russell Westbrook	2.755957
4	5.0	LeBron James	2.656892

```
Season 2016
```

	Rank	Player	MVP_Score
426	1.0	Russell Westbrook	3.498955
427	2.0	James Harden	3.252594
432	3.0	LeBron James	2.734909
445	4.0	Giannis Antetokounmpo	2.529146
429	5.0	Anthony Davis	2.413830

```
Season 2017
```

	Rank	Player	MVP_Score
861	1.0	LeBron James	3.279594
859	2.0	James Harden	3.215927
866	3.0	Russell Westbrook	2.861610
860	4.0	Anthony Davis	2.709005
862	5.0	Giannis Antetokounmpo	2.598780

```
Season 2018
```

	Rank	Player	MVP_Score
1340	1.0	James Harden	3.461875
1342	2.0	Giannis Antetokounmpo	2.939773
1349	3.0	Anthony Davis	2.480633
1341	4.0	Paul George	2.417262
1358	5.0	Russell Westbrook	2.399081

Season 2019

	Rank	Player	MVP_Score
1784	1.0	James Harden	3.342981
1788	2.0	Giannis Antetokounmpo	2.990547
1797	3.0	LeBron James	2.758400
1789	4.0	Luka Dončić	2.677236
1786	5.0	Damian Lillard	2.647054

Season 2020

	Rank	Player	MVP_Score
2263	1.0	Nikola Jokić	3.302264
2257	2.0	Giannis Antetokounmpo	2.677772
2253	3.0	Stephen Curry	2.515390
2258	4.0	Luka Dončić	2.458161
2255	5.0	Damian Lillard	2.250967

Season 2021

	Rank	Player	MVP_Score
2723	1.0	Nikola Jokić	3.667568
2716	2.0	Giannis Antetokounmpo	3.116663
2714	3.0	Joel Embiid	2.868197
2718	4.0	Luka Dončić	2.755121
2715	5.0	LeBron James	2.611073

Season 2022

	Rank	Player	MVP_Score
3244	1.0	Nikola Jokić	3.248749
3223	2.0	Luka Dončić	2.813935
3222	3.0	Joel Embiid	2.785803
3226	4.0	Giannis Antetokounmpo	2.471784
3225	5.0	Shai Gilgeous-Alexander	2.401524

Season 2023

	Rank	Player	MVP_Score
3702	1.0	Nikola Jokić	3.470047
3692	2.0	Luka Dončić	3.230572
3693	3.0	Giannis Antetokounmpo	2.901024
3694	4.0	Shai Gilgeous-Alexander	2.682363
3691	5.0	Joel Embiid	2.670856

Season 2024

	Rank	Player	MVP_Score
4187	1.0	Nikola Jokić	3.893210
4185	2.0	Shai Gilgeous-Alexander	3.501410
4188	3.0	Giannis Antetokounmpo	2.861473
4194	4.0	James Harden	2.477778
4186	5.0	Anthony Edwards	2.324009

Season 2025

	Rank	Player	MVP_Score
4680	1.0	Nikola Jokić	3.674747

	Rank	Player	MVP_Score
4673	2.0	Luka Dončić	3.003898
4674	3.0	Shai Gilgeous-Alexander	2.896978
4689	4.0	Victor Wembanyama	2.557042
4677	5.0	Tyrese Maxey	2.243469

The peformance of the created MVP metric was evaluated by comparing the model's top five players each season to the actual top five MVP finalists. On average, the model was able to correctly identify 3.6 of the 5 finalists oer season, which shows a good overall performance in capturing MVP level players.

Compare to real results

```
mvp_finalists = {
  2025: [],
  2024: ["Shai Gilgeous-Alexander", "Nikola Jokić", "Giannis Antetokounmpo", "Jayson Tatum",
  2023: ["Nikola Jokić", "Shai Gilgeous-Alexander", "Luka Dončić", "Giannis Antetokounmpo",
  2022: ["Joel Embiid", "Nikola Jokić", "Giannis Antetokounmpo", "Jayson Tatum", "Shai Gilgeo
  2021: ["Nikola Jokić", "Joel Embiid", "Giannis Antetokounmpo", "Devin Booker", "Luka Dončić
  2020: ["Nikola Jokić", "Joel Embiid", "Stephen Curry", "Giannis Antetokounmpo", "Chris Paul
  2019: ["Giannis Antetokounmpo", "LeBron James", "James Harden", "Luka Dončić", "Kawhi Leona
  2018: ["Giannis Antetokounmpo", "James Harden", "Paul George", "Nikola Jokić", "Stephen Cu
  2017: ["James Harden", "LeBron James", "Anthony Davis", "Damian Lillard", "Russell Westbro
  2016: ["Russell Westbrook", "James Harden", "Kawhi Leonard", "LeBron James", "Isaiah Thomas
  2015: ["Stephen Curry", "Kawhi Leonard", "LeBron James", "Russell Westbrook", "Kevin Durant
}

#compare results to actual finalists
results = []

for season, real_top5 in mvp_finalists.items():
  #skip 25-26 season as there are not results yet
  if season == 2025:
    continue

  model_top5 = set(
    final_nba[
      (final_nba["Season"] == season) &
      (final_nba["Rank"] <= 5)
```

```
    ]["Player"]
  )

  real_top5 = set(real_top5)
  overlap = len(model_top5 & real_top5)

  results.append({
    "Season" : season,
    "Model_Top5": list(model_top5),
    "Actual_Top5" : list(real_top5),
    "Overlap" : overlap,
    "Accuracy" : overlap/5
  })

results_df = pd.DataFrame(results).sort_values("Season")
results_df["Overlap"].mean()
```

```
np.float64(3.6)
```

In addition to identifying the top 5 candidates, the model was able to correctly predict the MVP winner in 50% of the seasons analyzed. This shows that while the model is effective at recognzining the top performers, is is less reliable at distinguising between closely ranked candidiates.

Check against actual winner

```
# Check for predicted winners
actual_mvp_winners = {
  2024: "Shai Gilgeous-Alexander",
  2023: "Nikola Jokić",
  2022: "Joel Embiid",
  2021: "Nikola Jokić",
  2020: "Nikola Jokić",
  2019: "Giannis Antetokounmpo",
  2018: "Giannis Antetokounmpo",
  2017: "James Harden",
  2016: "Russell Westbrook",
  2015: "Stephen Curry"
}
```

```
winner_results = []

for season, actual_winner in actual_mvp_winners.items():
  predicted_winner = final_nba[(final_nba["Season"] == season) & (final_nba["Rank"] == 1)]["

  winner_results.append({
    "Season": season,
    "Predicted_Winner": predicted_winner,
    "Actual_Winner": actual_winner,
    "Correct": predicted_winner == actual_winner
  })

winner_results_df = pd.DataFrame(winner_results).sort_values("Season")
winner_results_df

winner_results_df["Correct"].mean()
```

```
np.float64(0.5)
```

Additionally, the actual MVP winner appeared within the model's top two ranked players in 90% of the seasons analyzed. This demonstrates that the model is able to consistently idently the eventaul award winner in the top performers for the year. While the mettric may not always select the exact winner, the model is capable of narrowing down the MVP field significantly.

Comparing against top 2 finalists

```
#Check if predicted winner was in top 2
actual_top2 = {
  2024: ["Shai Gilgeous-Alexander", "Nikola Jokić"],
  2023: ["Nikola Jokić", "Shai Gilgeous-Alexander"],
  2022: ["Joel Embiid", "Nikola Jokić"],
  2021: ["Nikola Jokić", "Joel Embiid"],
  2020: ["Nikola Jokić", "Joel Embiid"],
  2019: ["Giannis Antetokounmpo", "LeBron James"],
  2018: ["Giannis Antetokounmpo", "James Harden"],
  2017: ["James Harden", "LeBron James"],
  2016: ["Russell Westbrook", "James Harden"],
  2015: ["Stephen Curry", "Kawhi Leonard"]
```

```
}

top2_results = []

for season, real_top2 in actual_top2.items():
  predicted_winner = final_nba[(final_nba["Season"] == season) & (final_nba["Rank"] == 1)]["
  
  top2_results.append({
    "Season": season,
    "Predicted_Winner": predicted_winner,
    "Actual_Top2": real_top2,
    "In_Actual_Top2": predicted_winner in real_top2
  })

top2_results_df = pd.DataFrame(top2_results).sort_values("Season")
top2_results_df

top2_results_df["In_Actual_Top2"].mean()
```

```
np.float64(0.9)
```

Overall, these various results suggest that the variable used and the weights assumed are able to account for many of the key statstical factors associated with MVP voting. However, the percentage of inaccuracies is able to highlight the influence of non-statsitcal factors discussed earlier, such as voter fatigue, team success and player narrative.

This study set out to evaluate whether a truly quanitiative model could effectively identify NBA MVP-level players using a wieghted combination of various performance metrics. By incorporting metrics that look at scoring, playmaking, impact, and defense, the constructed MVP metric is able to capture the key elements that influence MVP voting.

The results show that the model is very effective at identifying high level players, it correclty matched an average of 3.6 out of 5 MVP finalists per seaosn. Additioanlly, the model was able to successfully predict the MVP winner in 50% of the seasons analyzed and was able to put the eventual winner wihtin its top two ranking 90% of the time. This suggests that statistical performance does play a major role in MVP selection.

However, it is important to note that the mdoel did not perfectly replicate the MVP voting outcomes which highlight the influecne of subjective factors that can't be caputred by statistical data. Factors such as voter fatigue, team success, and plater narratve likely contrinute to the model's discrepancies compared to actual MVP resutls. This reinforces that idea that while statiscal metrics can privdoe valubale insight into a player's performce, it cannot fully account for the human element invlvoled in the MVP voting process.

There are several places for imporvement in a future project. First off, is looking it using more seasons and going back through possibly the enitre history of the NBA. Secondly, there are adiditonal variabels to look at such a team record or player efficiecny ratings (PER) that could be incorporated to enhance the model. Furthermore, a more complex method, like machine leaerning techniques, could be used to determine more optimal weights rahter than using the manually selected ones. Another option would be looking at if the model could predict the next seasons MVP winner based on the preivous years data alone.